Technical validation report

Results and validation method

Updated: 2026-10-20

1. Validation

1.1 Syllable counting

Radar bundles a local linguistic reference of common words. When a word is present, it uses that count; otherwise it applies spelling rules. Execution is local and does not depend on a connection or additional installations.

The reference used is CMU Pronouncing Dictionary (CMUdict; Carnegie Mellon University, 2024).

Fallback rules use documented English spelling conventions; the primary computational reference is CMUdict.

Measure	Result
Selected words	5.000
Comparable words	4.979
Excluded words	21
Local reference coverage	99,580 %
Exact agreement	100,000 %
Mean absolute error per word	0,000000

These results verify coverage and reproduction of the integrated reference. They do not replace an independent assessment of every possible pronunciation, website-content extraction, or human comprehension.

1.2 Comparison with reference software

The 100 English texts (50 easy and 50 difficult) were compared with Microsoft Word 16.0 using Flesch Reading Ease (Flesch, 1948) and English (United States). The measurements came from Word's readability statistics (Microsoft, n.d.). The original Radar scale preserves negative values; the bounded 0-100 diagnostic is reported separately to avoid boundary artefacts.

Measure	Result
Texts compared	100
Mean absolute difference, original scale	15,618
Pearson correlation, original scale	0,961
Lin concordance, original scale	0,845
Mean absolute difference, 0-100 diagnostic	0,839
Lin concordance, 0-100 diagnostic	0,999
Category agreement	99,0 %

2. Reproducibility

Detailed results, texts, and comparison columns are available in validation-results.xlsx. The workbook contains a summary and separate syllable and index sheets for both languages.

3. References

Speer, R. (2023). wordfreq (Version 3.1.1) [Computer software].

The Carnegie Mellon University. (2024). The CMU Pronouncing Dictionary.

Microsoft Corporation. (2026). Word Readability Statistics: Flesch Reading Ease.